



DBTalk

从 Redis 到 Valkey

· 开源社区如何快速创新 ·

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Valkey core team member



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加入 Valkey

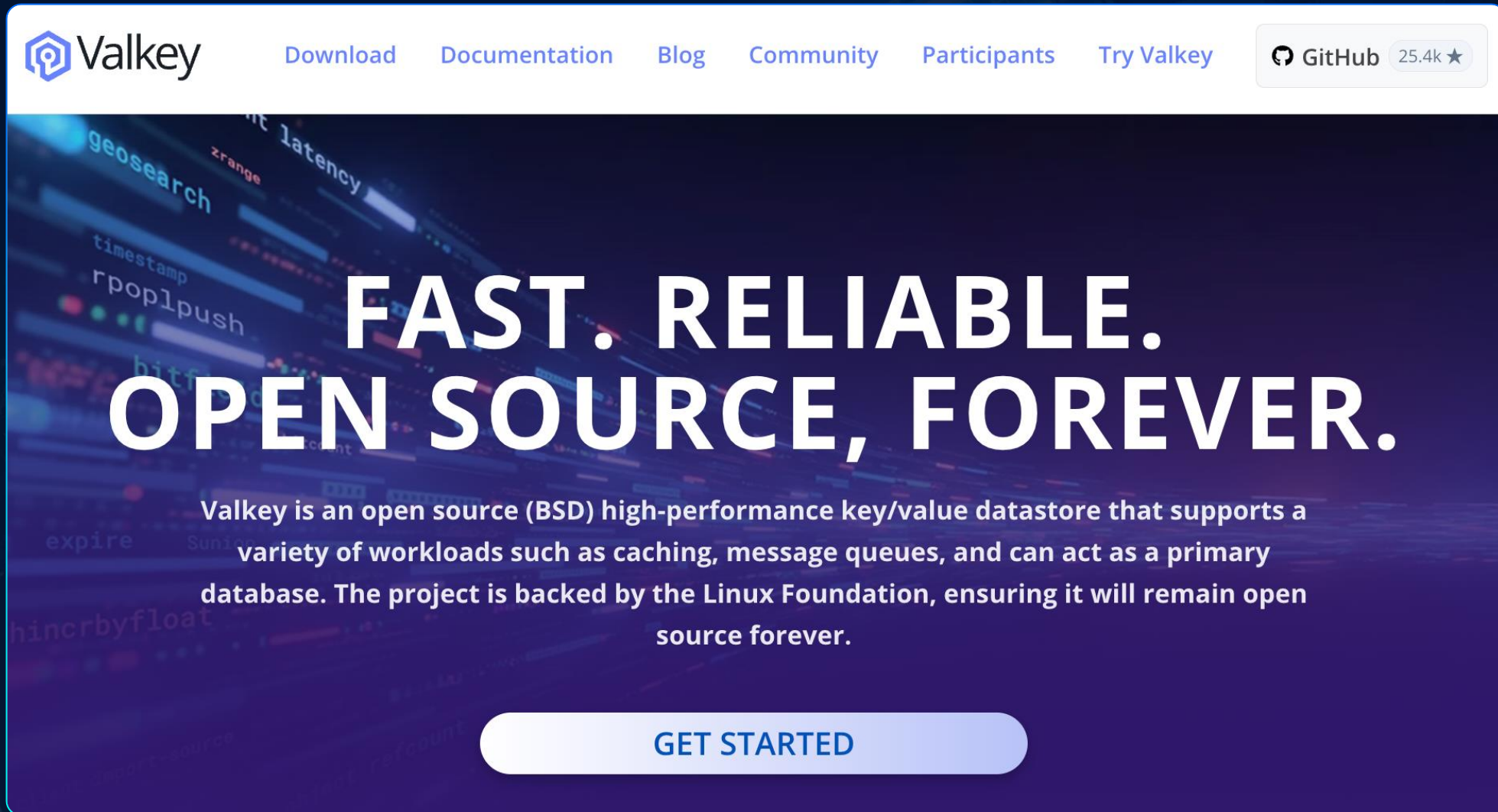
01

重新定义起点：Valkey 是什么？

Valkey Background



What is Valkey?

A screenshot of the Valkey website. The header features the Valkey logo on the left and navigation links for Download, Documentation, Blog, Community, Participants, and Try Valkey in the center. On the right, there is a GitHub repository link showing 25.4k stars. The main content area has a dark blue background with a blurred image of a server rack. Large white text reads "FAST. RELIABLE. OPEN SOURCE, FOREVER." Below this, a paragraph describes Valkey as an open source (BSD) high-performance key/value datastore. At the bottom, there is a white "GET STARTED" button.

Valkey

[Download](#) [Documentation](#) [Blog](#) [Community](#) [Participants](#) [Try Valkey](#)

[GitHub](#) 25.4k ★

FAST. RELIABLE. OPEN SOURCE, FOREVER.

Valkey is an open source (BSD) high-performance key/value datastore that supports a variety of workloads such as caching, message queues, and can act as a primary database. The project is backed by the Linux Foundation, ensuring it will remain open source forever.

[GET STARTED](#)

Valkey operating model

The 3-Clause BSD License

SPDX short identifier: BSD-3-Clause



open source
initiative
Approved License®

BSD-3 许可证

沿用 BSD-3 许可证，基本上你可以用它来任何事情。你之前拿 Redis 做什么，现在就能拿 Valkey 来做什么。



Linux 开源基金会

Valkey 项目是 Linux 开源基金会项目，厂商中立，不会被单一公司控制，确保 Valkey 永久保持开源。



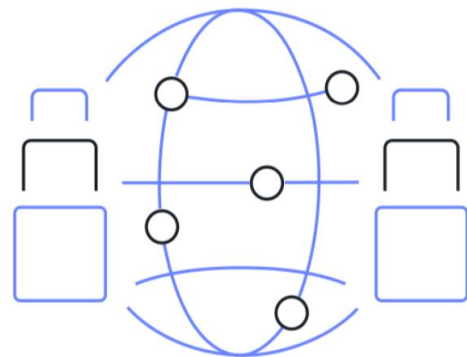
core-team

About

The core team, responsible for major decisions

TSC 技术指导委员会

项目由 TSC 技术指导委员会管理，委员会成员基于长期参与和贡献选出，重大事项投票决定，充分采纳意见。



自由开放透明

所有代码变更、讨论、提案都在 GitHub 上进行，任何人都可参与。定期在线会议，会议和录像对所有人公开。

Valkey TSC

THE BIRTH OF VALKEY

Many contributors, including companies providing hosted Redis-derived or Redis-compatible database services, just like Redis Ltd. does, have been using and contributing to Redis just as long as Redis Ltd. has existed. It was therefore an easy decision for many of them to continue the open source development as usual under the BSD license.

As Redis Ltd. owns the trademark rights to the name Redis, the open source project needed to continue under a different name. A group of six active contributors (one from each of Alibaba, Amazon, Ericsson, Google, Huawei and Tencent) with the support from several other companies launched Valkey as a Linux Foundation project. It was announced in a [press release](#) only eight days after Redis' license change. Three weeks later, a [second press release](#) announced seven more companies joining and the first release, Valkey 7.2.5.



Valkey TSC

Current Maintainers

Maintainers listed in alphabetical order by their github ID.

Maintainer	GitHub ID	Affiliation
Zhu Binbin	enjoy-binbin	Tencent
Wen Hui	hwware	Huawei
Madelyn Olson	madolson	Amazon
Ping Xie	pingxie	Google
Zhao Zhao	soloestoy	Alibaba
Viktor Söderqvist	zuiderkwast	Ericsson

<https://github.com/valkey-io/valkey/blob/8.0/MAINTAINERS.md>
<https://github.com/valkey-io/valkey/blob/unstable/MAINTAINERS.md>

Committee Chair

- **Madelyn Olson** Term: March 28, 2024 – Present

Current Maintainers

Maintainers listed in alphabetical order by their github ID.

Maintainer	GitHub ID	Affiliation
Binbin Zhu	@enjoy-binbin	Tencent
Harkrishn Patro	@hpatro	Amazon
Lucas Yang	@lucasyonge	Percona
Madelyn Olson	@madolson	Amazon
Jacob Murphy	@murphyjacob4	Google
Ping Xie	@pingxie	Oracle
Ran Shidlansik	@ranshid	Amazon
Zhao Zhao	@soloestoy	Alibaba
Viktor Söderqvist	@zuiderkwast	Ericsson

Valkey Governance

Valkey 项目采用 TSC (Technical Steering Committee) 技术指导委员会为核心的治理模式。强调技术领导和社区共识，TSC 负责项目的技术方向、重大决策和日常管理，所有决策通过共识优先，投票保障的机制运行。

Valkey 是社区驱动的开源项目，它不是单一主体的 Fork，而是一个社区 Fork，在尽最大努力保持社区开放。

不可有超过 1/3 的 TSC 来自同一组织

重大决议需要获得 TSC 2/3 票数

Project Governance

The Valkey project is managed by a Technical Steering Committee (TSC) composed of the maintainers of the Valkey repository. The Valkey project includes all of the current and future repositories under the Valkey organization. Contributors are defined as individuals with write access to the code within a repository. Maintainers are defined as individuals with full access to a repository and own its governance. Both maintainers and contributors shall be clearly listed in the MAINTAINERS.md file in a given project's repository. Maintainers of other repositories within the Valkey project are not members of the TSC unless explicitly added.

Technical Steering Committee

The TSC is responsible for oversight of all technical, project, approval, and policy matters for Valkey.

The TSC members are listed in the [MAINTAINERS.md](#) file in the Valkey repository.

At any time, no more than one third (1/3) of the TSC members may be employees, contractors, or representatives of the same organization or affiliated organizations. For the purposes of this document, "organization" includes companies, corporations, universities, research institutes, non-profits, governmental institutions, and any of their subsidiaries or affiliates. If, at any time, the 1/3 organization limit is exceeded (for example, due to changes in employment, company acquisitions, or organizational affiliations), the TSC shall be notified as soon as possible. The TSC must promptly take action to restore compliance, which may include removing or reassigning members in accordance with the procedures outlined in the [Termination of Membership](#) section. The TSC shall strive to resolve the situation within 30 days of notification, and document the steps taken to restore compliance.

The TSC shall appoint a Chair responsible for organizing TSC meetings. If the TSC Chair is removed from the TSC for the Chair steps down from that role, it is the responsibility of the TSC to appoint a new Chair.

The TSC may, at its discretion, add or remove members who are not maintainers of the main Valkey repository. The TSC may, at its discretion, add or remove maintainers from other repositories within the Valkey project.

Voting

The TSC shall strive for all decisions to be made by consensus. While explicit agreement of the entire TSC is preferred, it is not required for consensus. Rather, the TSC shall determine consensus based on their good faith consideration of a number of factors, including the dominant view of the TSC and nature of support and objections. The TSC shall document evidence of consensus in accordance with these requirements. If consensus cannot be reached, the TSC shall make the decision by a vote.

A vote shall also be called when an issue or pull request is marked as a major decision, which are decisions that have a significant impact on the Valkey architecture or design.

Technical Major Decisions

Technical major decisions include:

- Fundamental changes to the Valkey core datastructures
- Adding a new data structure or API
- Changes that affect backward compatibility
- New user visible fields that need to be maintained
- Adding or removing a new external library such as a client or module to the project when it affects runtime behavior

Technical major decisions shall be approved by a simple majority vote whenever one can be obtained. If a simple majority cannot be reached within a two-week voting period, and no TSC member has voted against, the decision may instead be approved through explicit "+2" support from at least two TSC members, recorded on the relevant issue or pull request. If the pull request author or issue proposer is a TSC member, their +1 counts toward the +2. If any TSC member casts a negative vote, the decision must follow the simple majority voting process and cannot be approved through +2. Once a technical major decision has been approved through the +2 mechanism, any subsequent concerns shall be raised through a new major decision process; +2 approvals are not retracted directly.

Governance Major Decisions

Governance major decisions include:

- Adding TSC members or involuntary removal of TSC members
- Modifying this governance document
- Delegation of maintainership for projects or governance authority
- Creating, modifying, or removing roles within the Valkey project
- Any change that alters voting rules, TSC responsibilities, or project oversight
- Structural changes to the TSC, including composition limits

Governance major decisions shall require approval by a super-majority vote of at least two thirds (2/3) of the entire TSC.

Any member of the TSC can call a vote with reasonable notice to the TSC, setting out a discussion period and a separate voting period. Any discussion may be conducted in person or electronically by text, voice, or video. The discussion shall be open to the public, with the notable exception of discussions involving embargoed security issues or the addition or removal of maintainers, which will be private. In any vote, each voting TSC member will have one vote. The TSC shall give at least two weeks for all members to submit their vote. Except as specifically noted elsewhere in this document, decisions by vote require a simple majority vote of all voting members. If a vote results in a tie, the status quo is preserved. It is the responsibility of the TSC Chair to help facilitate the voting process as needed to make sure it completes within the voting period.

Termination of Membership

A maintainer's access (and accordingly, their position on the TSC) will be removed if any of the following occur:

- Involuntary Removal: Removal via the [Governance Major Decision](#) voting process.
- Resignation: Written notice of resignation to the TSC.
- Unreachable Member: If a member is unresponsive for more than six months, the remaining active members of the TSC may vote to remove the unreachable member by a simple majority.

Technical direction for other Valkey projects

The TSC may delegate decision-making for other projects within the Valkey organization to the maintainers responsible for those projects. Delegation of decision-making for a project is considered a [Governance Major Decision](#). Projects within the Valkey organization must indicate the individuals with commit permissions by updating the MAINTAINERS.md within their repositories.

The TSC may, at its discretion, overrule the decisions made by other projects within the Valkey organization, although they shall show restraint in doing so.

License of this document

This document may be used, modified, and/or distributed under the terms of the [Creative Commons Attribution 4.0 International \(CC-BY\) license](#).

02

深度对比：社区模式如何解决经典性能难题

案例一：I/O多线程的深度优化：从“够用”到“极致”



Redis 单线程串行执行多条命令

Main thread

GET KEY

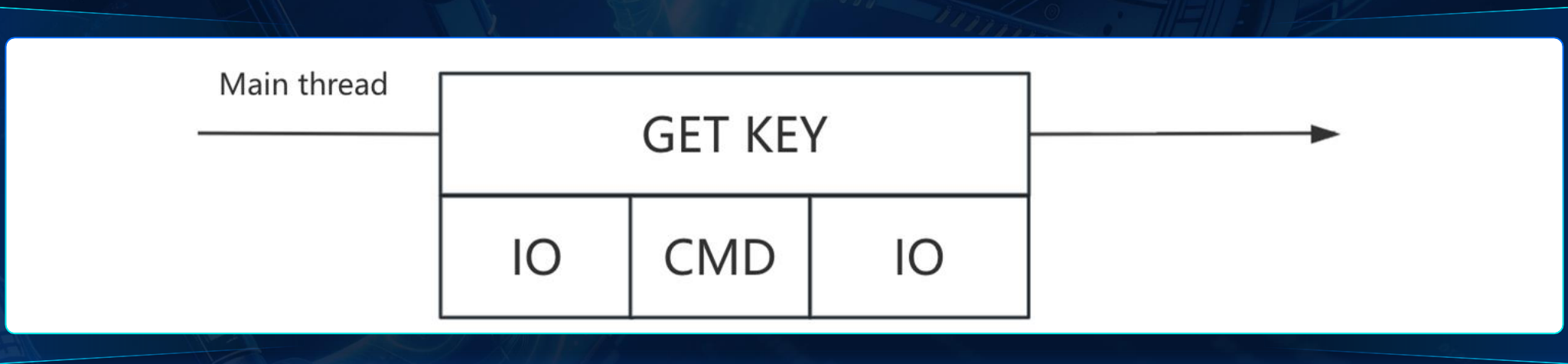
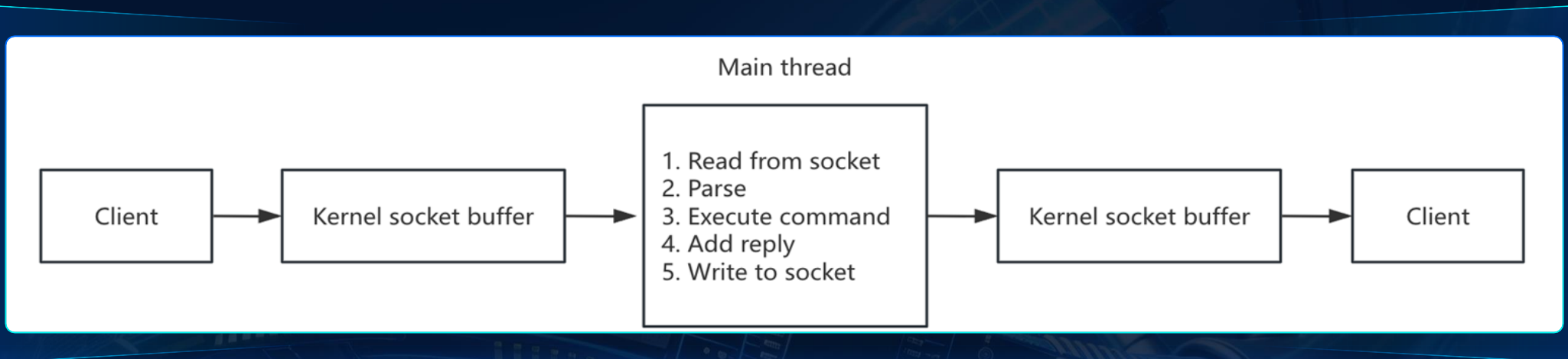
Main thread

GET KEY

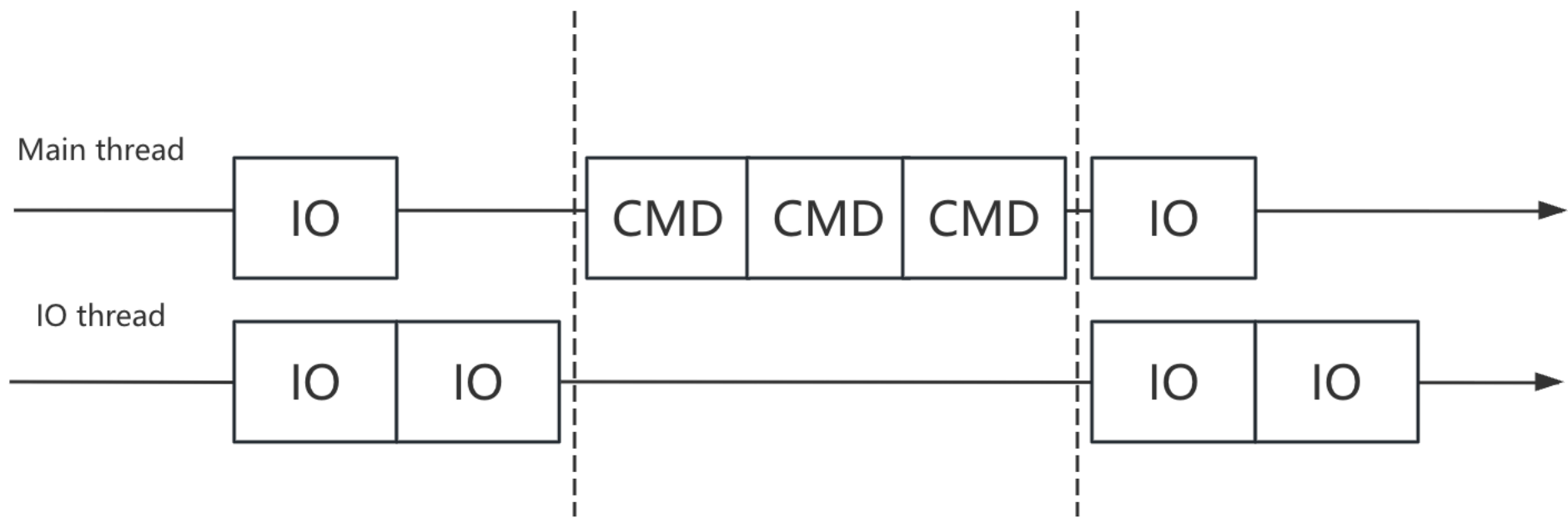
GET KEY2

GET KEY3

Redis 执行命令拆分



Redis 6.0 引入 IO 多线程



Redis issue - Threaded IO improvements (handle TLS)

Threaded IO improvements (handle TLS) #11119

✓ Closed



oranagra opened on Aug 14, 2022 · edited by oranagra

Edits ▾

Member



A serious redesign of the threaded IO is needed.
Most importantly in order to support TLS so that the encryption work can be offloaded to the treads.
Possibly to assign clients for threads rather than the current fan-out approach.
TBD.



Redis issue - Reaching 1 million requests per second

Reaching 1 million requests per second on a single Redis instance #12489

Open



toutou-dan opened on Aug 15, 2023 · edited by madolson

Edits ...

The following is a proposal to accelerate Redis performance by:

1. Improving io-thread efficiency by totally offloading network layer from main thread.
2. Reducing load on main thread remaining functionality by moving memory management and response formatting to the io-threads.
3. Amortizing memory access latency by batching dictionary searches.

We believe than by implementing these steps, a single Redis instance running on a multicore machine will deliver more than 1 million rps, up from under 400K rps as it is today.

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

Valkey issue - Reaching 1 million requests per second

[NEW] Reaching 1 million requests per second on a single Valkey instance #22

✓ Closed



PingXie opened on Mar 25, 2024 · edited by enjoy-binbin

Edits ▾

Member



Summary: This problem was solved by the following three PRs.

1. [Async IO threads #758](#)
2. [Io thread work offload #763](#)
3. [Improve multithreaded performance with memory prefetching #861](#)

More follow-up are

- [Followup items from #758 Async IO threads #761](#)

This is a true gem :).

The proposal was originally brought up by [@toutou-dan](#) and below is a verbatim copy of [redis/redis#12489](#)

Valkey I/O threading meeting

Uri 的屏幕

binbinzhu

Viktor

Ping Xie

Madelyn Olson

Dan Toulou

Uri

PowerPoint

Edit View Window Help

PowerPoint Slide Show - [1mtps (1)]

Mon 6 May 17:27

Epoll Offload

```
graph TD
    subgraph Main_Thread [Main Thread]
        EPOLL_WAIT[EPOLL WAIT] --> Send_Read_Jobs[Send Read Jobs]
        Send_Read_Jobs --> Prefetch_Commands_Keys[Prefetch Commands Keys]
        Prefetch_Commands_Keys --> Process_Commands[Process Commands]
        Process_Commands --> Send_Write_Free_Jobs[Send Write & Free Jobs]
        Send_Write_Free_Jobs --> IO_PENDING_JOBS{IO PENDING JOBS?}
        IO_PENDING_JOBS -- NO --> EPOLL_WAIT
        IO_PENDING_JOBS -- YES --> Send_EPOLL_Job[Send EPOLL Job]
        Send_EPOLL_Job --> Send_Read_Jobs
    end

    subgraph IO_Thread_1 [IO Thread 1]
        Wait_for_Jobs1((Wait for Jobs)) --> READ_Command_Lookup1[READ + Command Lookup]
        READ_Command_Lookup1 --> WRITE1[WRITE]
        WRITE1 --> Free_Objects1[Free Objects]
        Free_Objects1 --> Wait_for_Jobs1
    end

    subgraph IO_Thread_2 [IO Thread 2]
        Wait_for_Jobs2((Wait for Jobs)) --> READ_Command_Lookup2[READ + Command Lookup]
        READ_Command_Lookup2 --> WRITE2[WRITE]
        WRITE2 --> Free_Objects2[Free Objects]
        Free_Objects2 --> Wait_for_Jobs2
    end

    Send_Write_Free_Jobs --> Wait_for_Jobs1
    Send_Write_Free_Jobs --> Wait_for_Jobs2
    Send_Write_Free_Jobs --> EPOLL_Non_block[EPOLL Non block]
```

转写文字

U Uri

22:26:45
And this is the current outside. So, so, the invocation most of the time. And the, the main city is still doing 30%.

22:26:47
这就是外面的电流。是的，这大多数时候都是这句话。而且，主要城市仍然占到了30%。

22:26:56
And. So this is what I suggest in the 1st PR. The main series is, is the nipple weight and set the wheat jobs and asynchronously it continues to process command.

22:26:59
还有。所以，我在第一篇公文中建议如下。主要系列是，是nipple重量和设置小麦作业，然后异步地继续处理命令。

22:27:09
Then you send right and. And I also said this all the time waiting for new jobs.

22:27:09
然后你向右发送。我在等待新工作的时候也经常这么说。

22:27:16
And so that's the 1st view. Second PR. I am. We show that the is April weight costing 8% after we off route?

22:27:16
这就是我们的第一个观点。第二次公关。我是。我们发现，在我们离开航线后，4月份的重量是否下降了8%？

22:27:28
All the audio to switch. So. So what we did is. We check

22:27:29
所有的音频都需要切换。是的。所以我们

保存转写文字

Valkey I/O threading PRs, released in Valkey 8.0

Async IO threads #758

Merged madolson merged 1 commit into valkey-io:unstable from uriyage:async-io-th

Conversation 5 Commits 1 Checks 0 Files changed 38



uriyage commented on Jul 8, 2024

Member ...

This PR is 1 of 3 PRs intended to achieve the goal of 1 million requests per second, as detailed by [dan toutou](#) in [#22](#). This PR modifies the IO threads to be fully asynchronous, which is a first and necessary step to allow more work offloading and better utilization of the IO threads.

Io thread work offload #763

Merged madolson merged 3 commits into valkey-io:unstable from uriyage:io-thread-work-

Conversation 68 Commits 3 Checks 0 Files changed 11



uriyage commented on Jul 9, 2024 • edited

Member ...

IO-Threads Work Offloading

This PR is the 2nd of 3 PRs intended to achieve the goal of 1M requests per second. (1st PR: [#758](#))

This PR offloads additional work to the I/O threads, beyond the current read-parse/write operations, to better utilize the I/O threads and reduce the load on the main thread.

Improve multithreaded performance with memory prefetching #861

Merged madolson merged 7 commits into valkey-io:unstable from uriyage:commands-prefetching on Aug 27, 2024

Conversation 100 Commits 7 Checks 0 Files changed 17



uriyage commented on Aug 1, 2024

Member ...

This PR is the last of three pull requests intended to achieve the goal of processing one million requests per second.

1st PR: [#758](#)

2nd PR: [#763](#)

With these 3 PRs, we can now handle up to 1.2 million requests per second, up from 200K without IO-threads and ~380K with the current IO threads implementation.

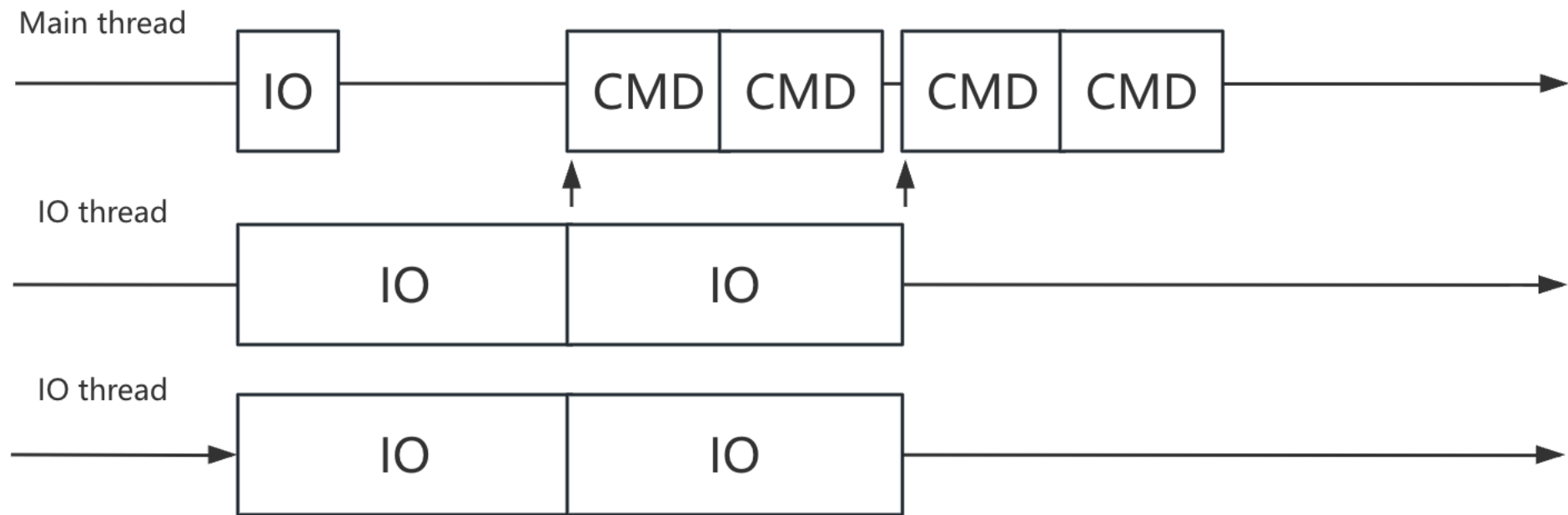
This PR utilizes the IO threads to execute commands in batches, allowing us to prefetch the dictionary data in advance.

<https://github.com/valkey-io/valkey/pull/758>

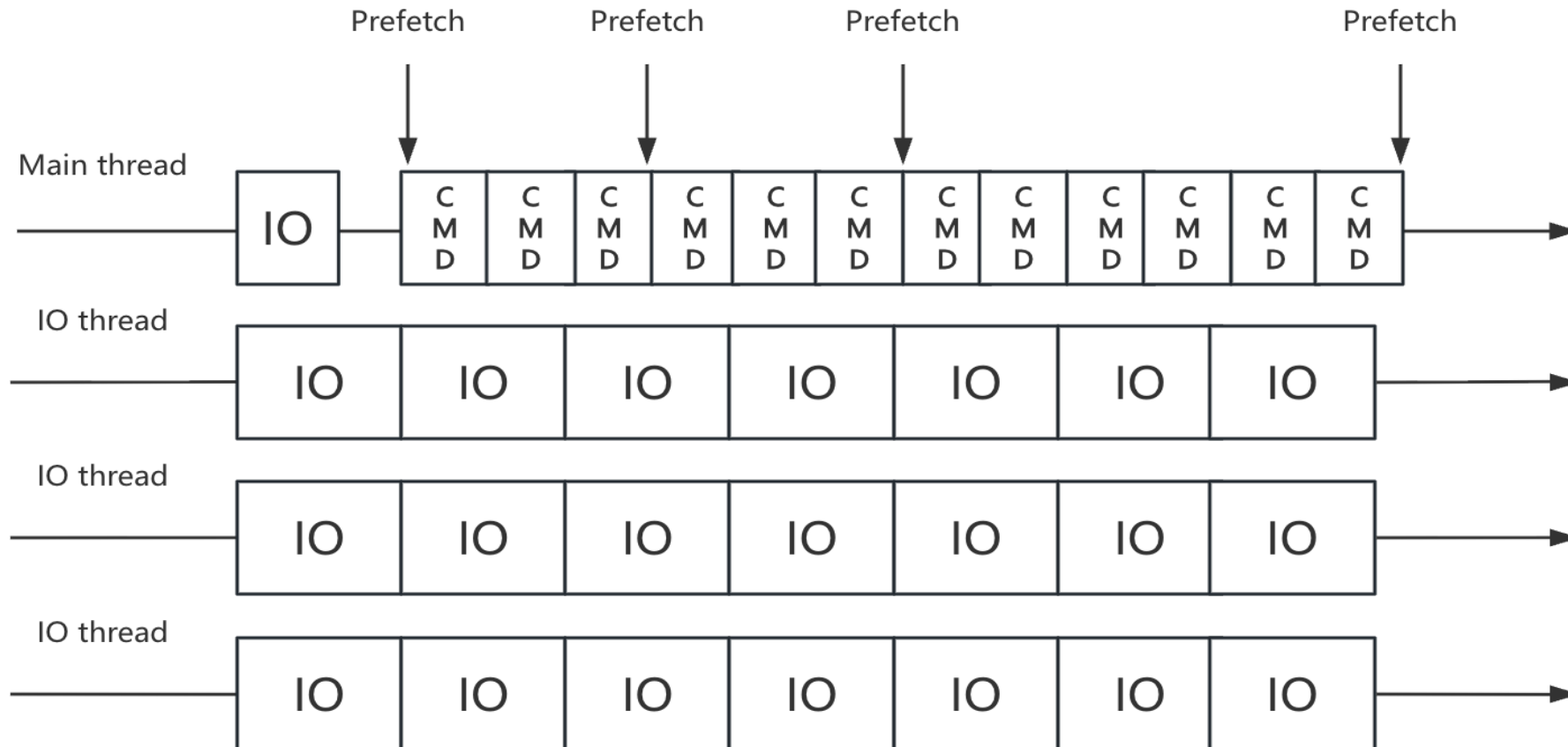
<https://github.com/valkey-io/valkey/pull/763>

<https://github.com/valkey-io/valkey/pull/861>

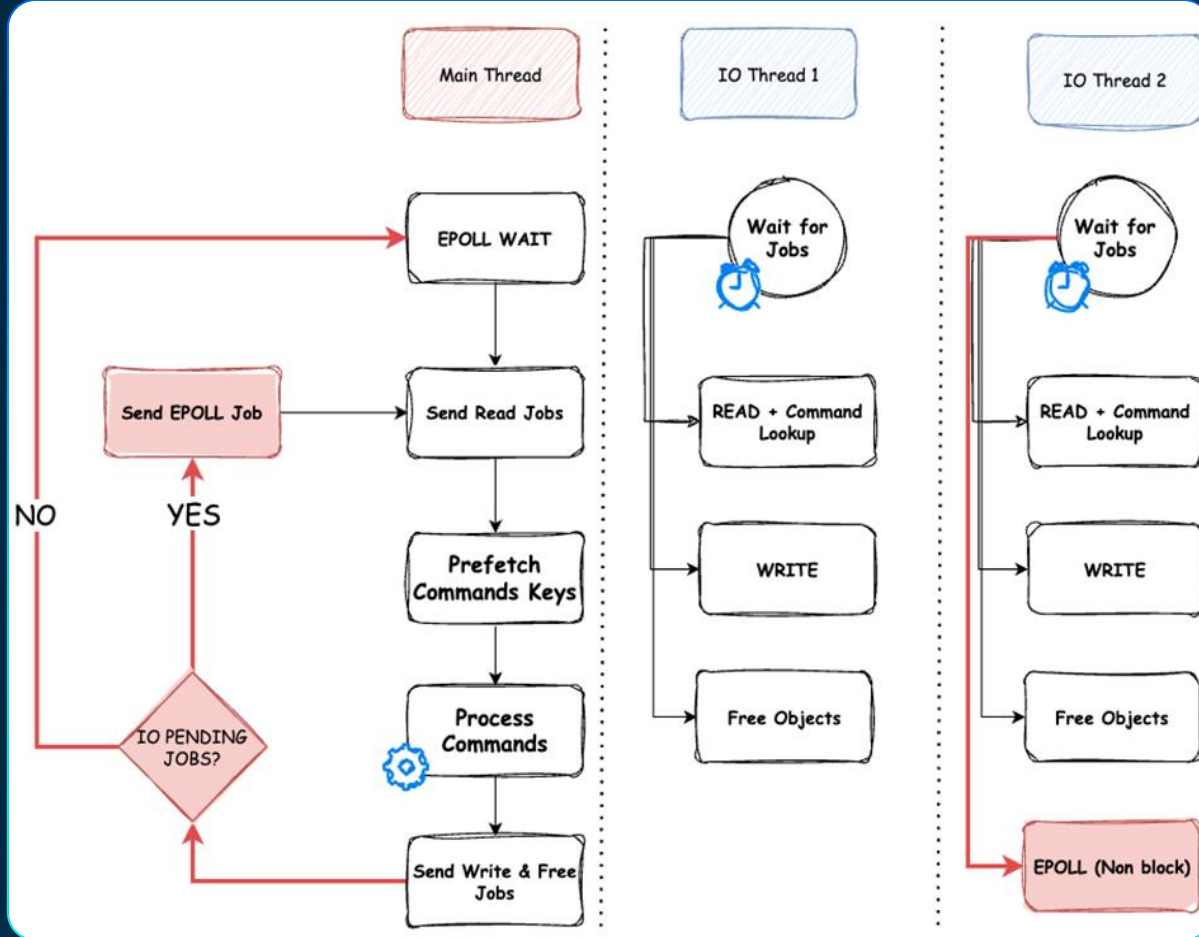
Valkey 8.0 Async IO threads



Valkey 8.0 Async IO threads

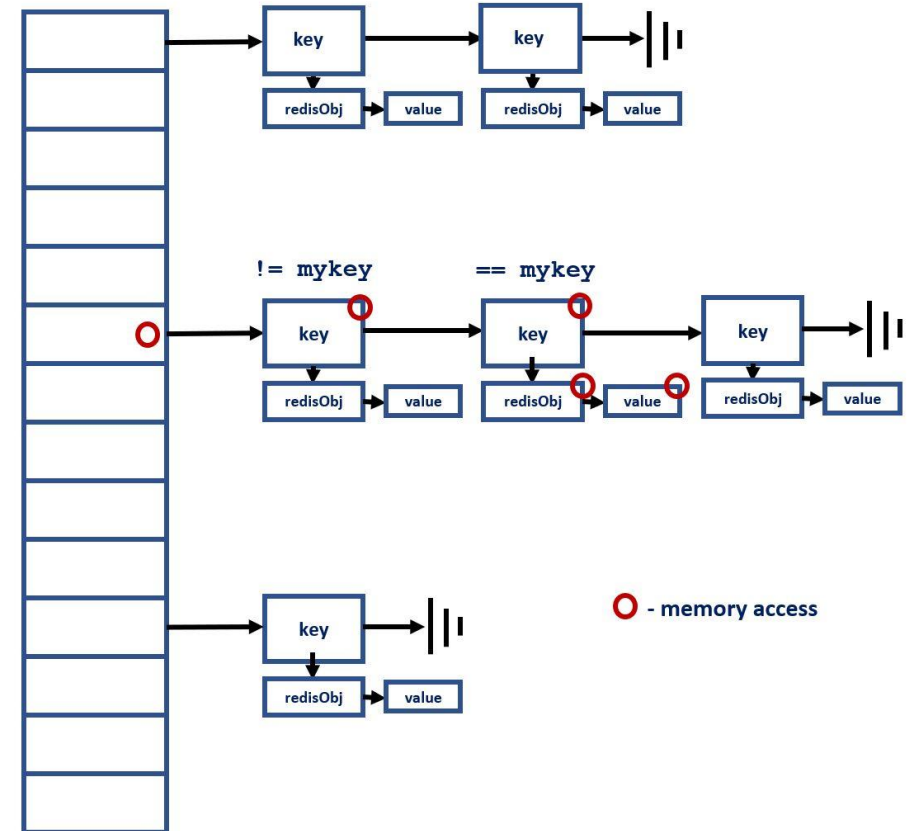


Valkey 8.0 Unlock 1 Million RPS



<https://valkey.io/blog/unlock-one-million-rps/>

SET mykey new

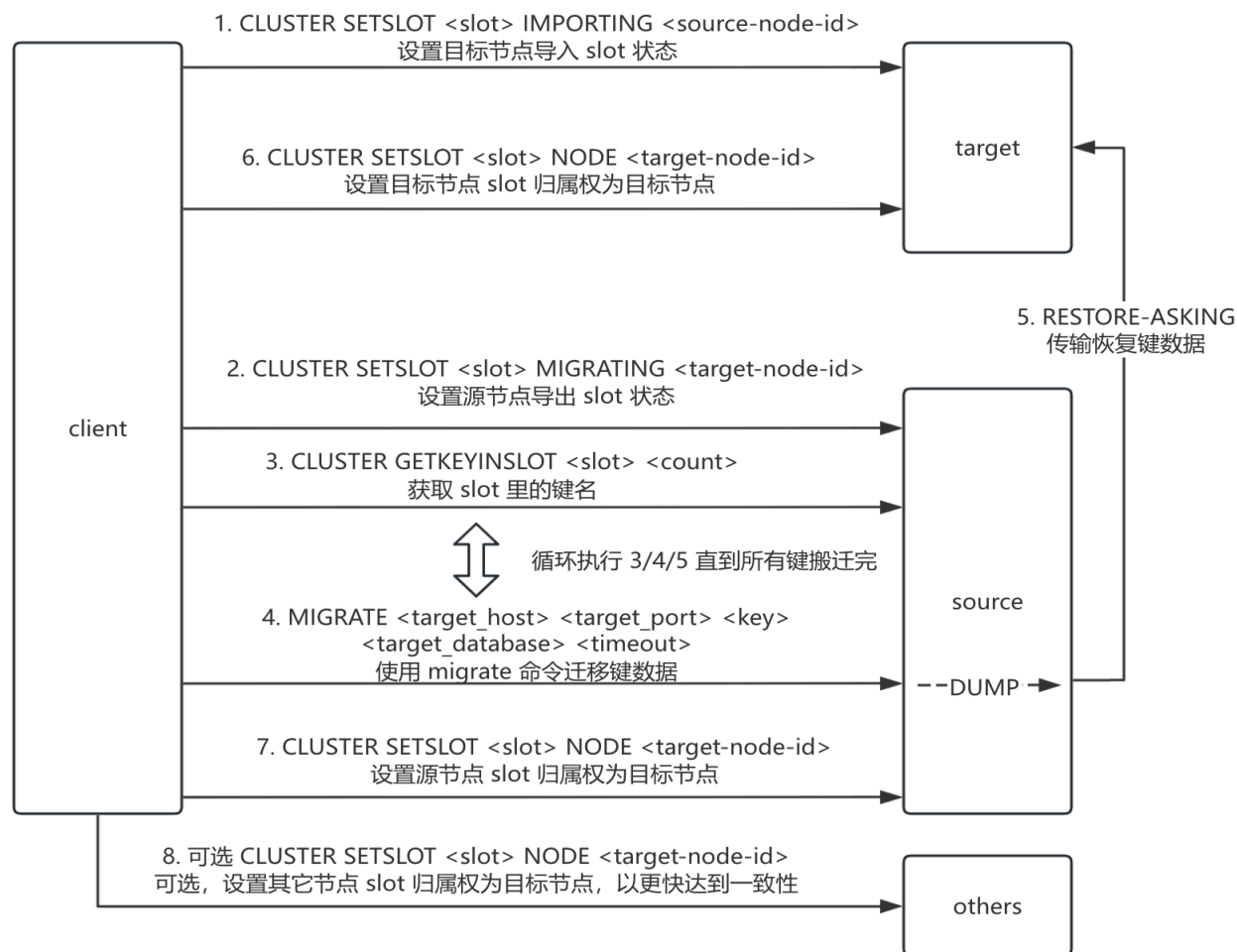


<https://valkey.io/blog/unlock-one-million-rps-part2/>

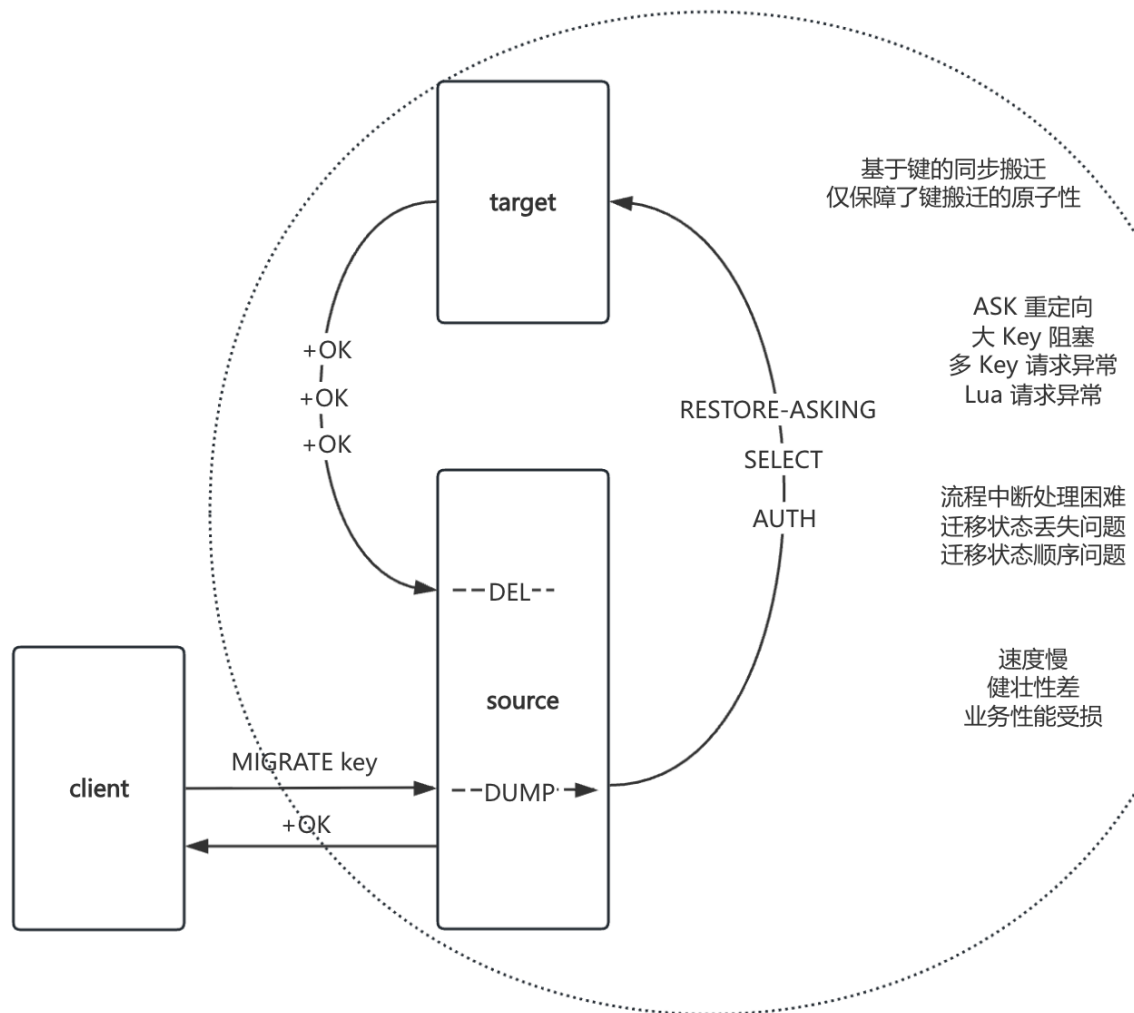
案例二：社区共治：水平扩缩容的加速



社区按键同步搬迁方案



按键同步迁移，基础但缓慢



影响业务请求 >>>

- ASK 重定向
- 大 Key 阻塞
- 多 Key 请求异常
- Lua 请求异常

健壮性差 >>>

- 迁移中断难处理
- 迁移状态丢失
- 迁移状态乱序

扩容速度慢 >>>

- 单线程受限
- 需限制 QPS 避免影响正常请求
- 周期过长无法快速响应

Redis issue - Atomic slot migration HLD

Atomic slot migration HLD #10933

New issue



Open



madolson opened on Jul 5, 2022 · edited by madolson

Edits ▾

Contributor



High level design

We want to introduce a new Slot Migration process that atomically migrates a set of slots from a source node to a target node.

We will implement a net new slot serialization strategy which emulates the behavior of BGSave but sends a series of RESTORE COMMANDS instead of the RDB snapshot serialization format. The strategy being that we fork when migrating a batch of slots, we establish the new connection, and then iterate over the slot space and sent the restore commands. Once all the data for a slot is migrated, the accumulated buffer for the slot is drained. Once the buffer is drained, we block all write commands for the slot while we do a coordinated handoff, this can be upgraded to 2PC for cluster v2, to complete the transaction and make sure the target node takes ownership of the slot. Once the target has confirmed it owns the slot, the source node will asynchronously purge its slot data. This process will be repeated for all slots being migrated, one at a time. The forked process will be used to migrate a set of slots, once this set of slots has been migrated, the fork process ends.

Assignees

No one assigned

Labels

No labels

Type

No type

Projects



Redis Backlog

Status

Backlog



Valkey issue - Atomic slot migration HLD

[NEW] Atomic slot migration HLD #23

✓ Closed

#1949



PingXie opened on Mar 25, 2024 · edited by enjoy-binbin

Edits ▾

Member



See [#1949](#) for the latest updates.

Problem Statement

Efficient slot migration is critical for maintaining the scalability and resilience of Valkey clusters, but the current client-driven migration approach presents significant challenges.

Operators currently need to perform multiple manual steps to migrate slots, including setting migration states, transferring data, and updating ownership, all of which are prone to errors and inefficiencies.

While existing migration methods ensure availability during the process, they rely on mechanisms like redirection (-MOVED, -ASK) to maintain client access to data. These mechanisms introduce significant client-side complexity, requiring clients to handle redirections, retries, and reconfigurations during migration. Ensuring atomicity and consistency during migrations remains a challenge, especially for operations involving multiple keys or large datasets.

Compounding these challenges, limited observability leaves operators without adequate insight into the migration process, making it difficult to monitor progress or resolve issues efficiently.

Valkey ATM 讨论 , Tencent Cloud 和 GCP 协作推进



PingXie on Dec 16, 2024 · edited by PingXie

Edits ▼ Member Author ...

This atomic slot migration proposal incorporates valuable input from offline discussions with [@enjoy-binbin](#) and [@murphyjacob4](#). It builds on the shared foundation of using RDB format for slot data transfer via a child process and the ownership transfer flow, while also addressing considerations discussed during the process, such as support for targets serving traffic, epoch bumps, and the trade-off between availability and durability.

[@valkey-io/core-team](#) PTAL

 2  1



murphyjacob4 commented on Jan 23, 2025

Member Author ...

Hi folks, so [@enjoy-binbin](#), [@PingXie](#), and I discussed offline. Tencent has offered a solution that they have developed internally. Moving forward, [@enjoy-binbin](#) and I will join efforts and work on bridging gaps in the Tencent solution to meet the requirements we outlined in [#23](#). Hopefully we will have a shared PR for review soon.

Given this, I am going to go ahead and close this pull request.

 1

Valkey Atomic Slot Migration meeting

会议

Madelyn Olson 的屏幕

binbinzhu

binbinzhu

Ran Shidlansik (Amaz...

Madelyn Olson

Jacob Murphy

Ping Xie

Viktor Söderqvist (Eric...

github.com

Introduce atomic slot migration by murphyjacob4 · Pull Request #1949 · valkey-io/valkey

Add RDB Version to "INF

Open

murphyjacob4 wants to merge 105 commits into `valkey-io:unstable` from `enjoy-binbin:slot_migration_murphyjacob4`

Conversation 328

Commits 105

Checks 1

Files changed 39

+5,133 -331



murphyjacob4 commented on Apr 13 · edited

Member

Introduces a new family of commands for migrating slots via replication. The procedure is driven by the source node which pushes an AOF formatted snapshot of the slots to the target, followed by a replication stream of changes on that slot (a manual failover).

This solution is an adaptation of the solution provided by [@enjoy-binbin](#), combined with the solution I previously posted at [#1591](#), modified to meet the designs we had outlined in [#23](#).

New commands

- `CLUSTER MIGRATE SLOTSRANGE start end [start end]... NODE node-id`: Begin sending the slot via replication to the target. Multiple targets can be specified by repeating `SLOTSRANGE ... NODE ...`
- `CLUSTER CANCELMIGRATION (NAME <name> | ALL)`: Cancel either one or all slot migrations
- `CLUSTER MIGRATIONS`: See a recent log of migrations

This PR only implements "one shot" semantics with an asynchronous model. Later, "two phase" (e.g. slot level replicate/failover commands) can be added with the same core.

Slot migration jobs

Reviewers

zuiderkwast

gmbnomis

PingXie

enjoy-binbin

madolson

Copilot

ranshid

hwware

At least 1 approving review is required to merge this pull request.

Still in progress? [Convert to draft](#)

Assignees

enjoy-binbin

Atomic slot migration, released in Valkey 9.0

Introduce atomic slot migration #1949

Merged

[madolson](#) merged 125 commits into [valkey-io:unstable](#) from [enjoy-binbin:slot_migration_murphyjacob4](#) on Aug 12, 2025

Conversation 519

Commits 125

Checks 63

Files changed 40

+5,147 -376



murphyjacob4 commented on Apr 13, 2025

Member

• edited by enjoy-binbin

Introduces a new family of commands for migrating slots via replication. The procedure is driven by the source node which pushes an AOF formatted snapshot of the slots to the target, followed by a replication stream of changes on that slot (a la manual failover).

This solution is an adaptation of the solution provided by [@enjoy-binbin](#), combined with the solution I previously posted at [#1591](#), modified to meet the designs we had outlined in [#23](#).

New commands

- `CLUSTER MIGRATESLOTS SLOTSRANGE start end [start end]... NODE node-id` : Begin sending the slot via replication to the target. Multiple targets can be specified by repeating `SLOTSRANGE ... NODE ...`
- `CLUSTER CANCELMIGRATION ALL` : Cancel all slot migrations
- `CLUSTER GETSLOTMIGRATIONS` : See a recent log of migrations

This PR only implements "one shot" semantics with an asynchronous model. Later, "two phase" (e.g. slot level replicate/failover commands) can be added with the same core.

Reviewers

 zuiderkwast

 enjoy-binbin

 madolson

 Copilot

 PingXie

+2 more reviewers

 gmbnomis

 hwware

Assignees

 enjoy-binbin

Labels

cluster

enhancement

major-decision-approved

needs-doc-pr

release-notes

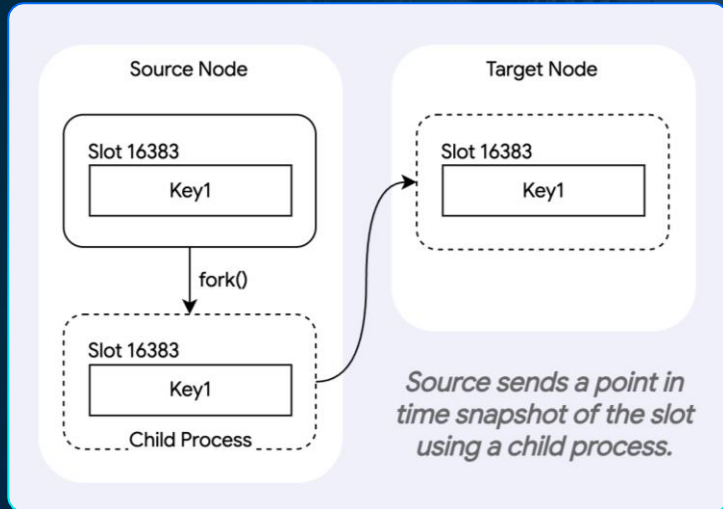
run-extra-tests

<https://github.com/valkey-io/valkey/pull/1949>

Valkey 9.0 Atomic Slot Migration

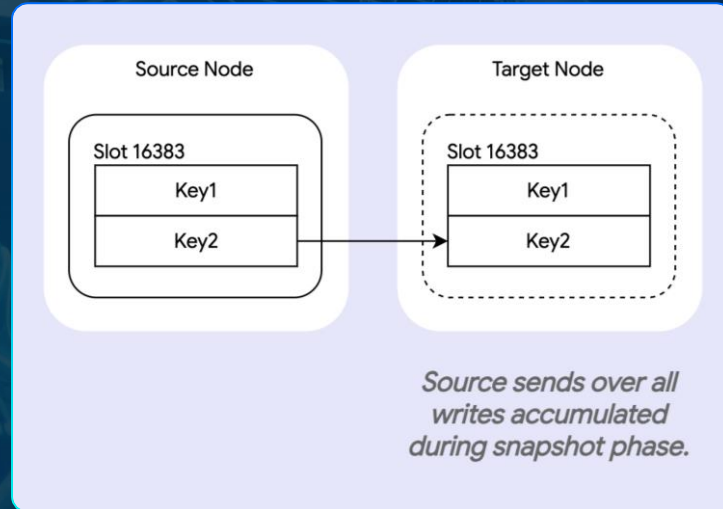
01

SNAPSHOT



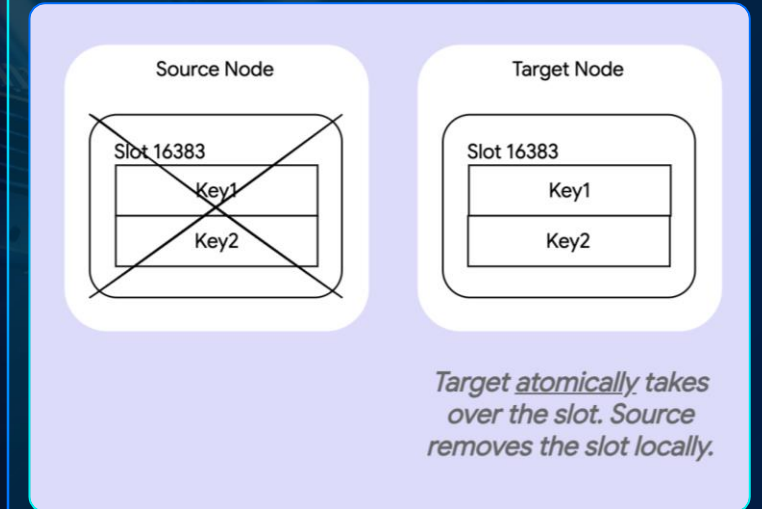
02

STREAM



03

FINALIZE



03

系统进化：由社区驱动的全方位卓越

| Valkey is not the open source version of Redis



Valkey != *Redis*

What is new in Valkey 8.0?

2024-09-16

Valkey 8.0.0



io-threads, 1M Requests per second (RPS)

Dual channel replication

Enhanced slot migration reliability

Memory efficiency

Valkey Over RDMA transport (experimental)

<https://valkey.io/blog/valkey-8-0-0-rc1/>
<https://github.com/valkey-io/valkey/blob/8.0/00-RELEASENOTES>

What is new in Valkey 8.1?

2024-09-16

Valkey 8.0.0

2025-03-31

Valkey 8.1.0

New hashtable, reduced memory overhead by 20%

valkey-bundle: JSON, Bloom, Search and LDAP modules

Observability: New command log feature

<https://valkey.io/blog/valkey-8-1-0-ga/>
<https://github.com/valkey-io/valkey/blob/8.1/00-RELEASENOTES>

What is new in Valkey 9.0?

2024-09-16

Valkey 8.0.0

2025-03-31

Valkey 8.1.0

2025/10/21

Valkey 9.0.0

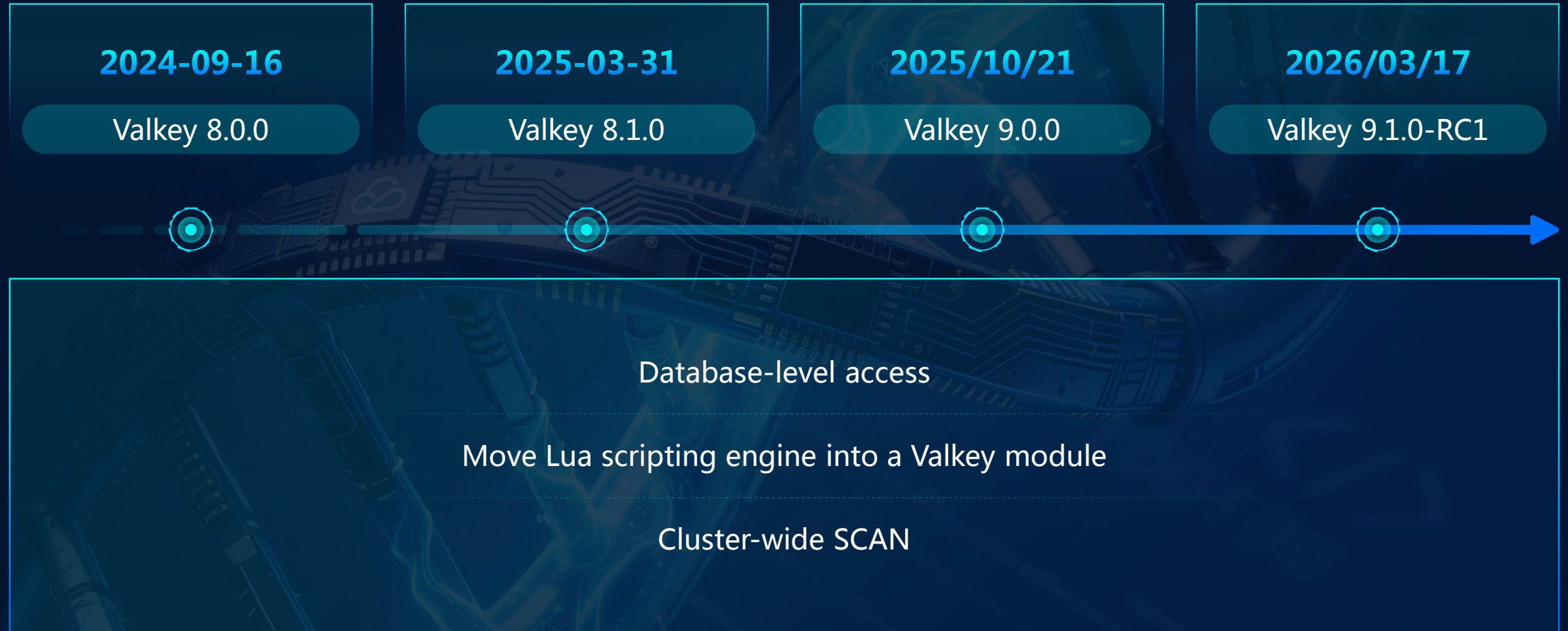
Multiple databases in cluster mode

Atomic slot migration

Expiration on hash field items

<https://valkey.io/blog/introducing-valkey-9/>
<https://github.com/valkey-io/valkey/blob/9.0/00-RELEASENOTES>

What is new in Valkey 9.1?



修复边界 BUG 引入的性能损失

Merged

Fix commandlog large-reply when using reply copy avoidance #2652

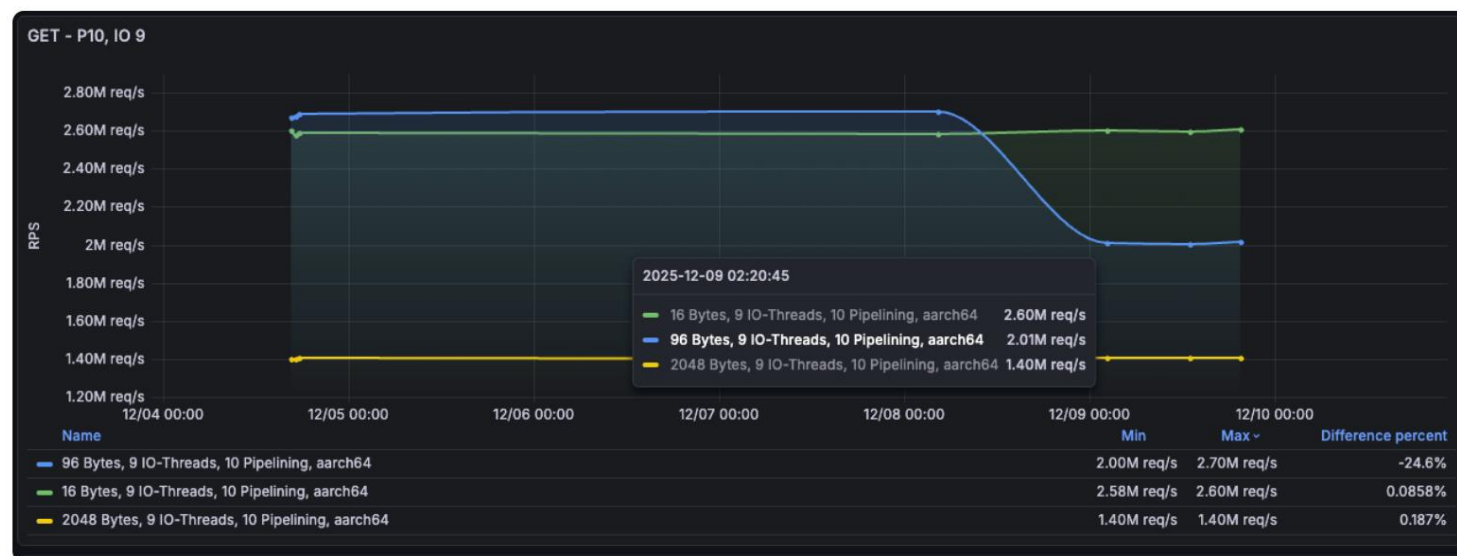
[enjoy-binbin](#) merged 2 commits into [valkey-io:unstable](#) from [enjoy-binbin:fix_large_reply](#)



roshkhatri commented on Dec 11, 2025 • edited

Member

I noticed that this commit is effecting the performance Get commands from 2.7M to 2.0M for 96 bytes data size, 9 IO-Threads and 10 Pipelining. this is about 25% drop



You can see more [here](#)

[@enjoy-binbin](#) Please can you take a look here

Valkey Performance Dashboards

Valkey Performance Dashboards

Valkey Performance Dashboards provide a consolidated view of throughput trends across versions, helping teams validate improvements and identify regressions quickly.

 **README**

 Code of conduct

 Contributing

 License

 Security



Performance monitoring

Valkey Performance Dashboards provide a consolidated view of throughput trends across versions, helping contributors validate improvements and identify regressions quickly.

- [Performance Overview](#) - Compare throughput across Valkey versions
- [Unstable Branch Dashboard](#) - Track performance of all commits in the unstable branch

<https://valkey.io/performance/>

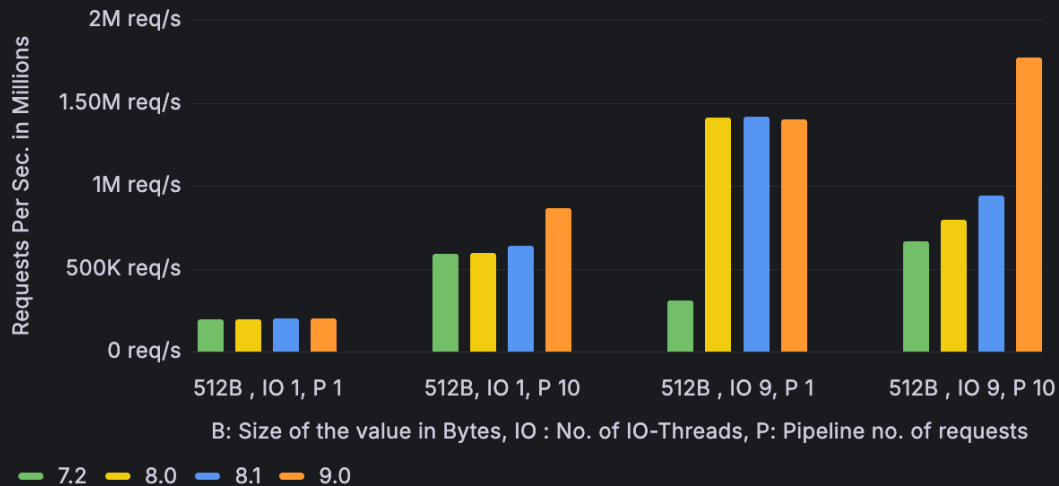
<https://github.com/valkey-io/valkey?tab=readme-ov-file#performance-monitoring>

Valkey Throughput Across Versions

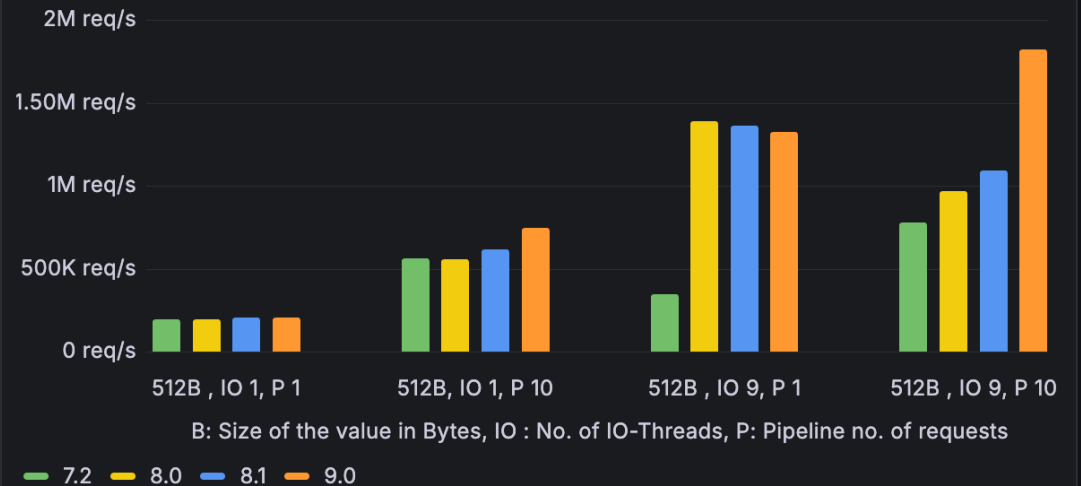


ValkeyThroughput Comparison

SET COMMAND ⓘ



GET COMMAND ⓘ



Valkey Unstable Branch Performance

Valkey Performance - Per Command

SET - P10, IO 9



Name	Min	Max ~	Difference percent
16 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	1.77M req/s	2.23M req/s	18.9%
96 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	1.71M req/s	2.23M req/s	25.9%
128 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	1.86M req/s	2.17M req/s	13.7%
2048 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	1.39M req/s	1.40M req/s	-0.154%

GET - P10, IO 9



Name	Min	Max ~	Difference percent
16 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	2.39M req/s	3.34M req/s	33.7%
96 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	1.97M req/s	3.28M req/s	63.8%
128 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	2.27M req/s	2.57M req/s	11.9%
2048 Bytes, 9 IO-Threads, 10 Pipelining, aarch64	1.40M req/s	1.41M req/s	0.158%

SET - P1, IO 9



Name	Min	Max ~	Difference percent
16 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.43M req/s	1.63M req/s	11.8%
96 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.44M req/s	1.58M req/s	7.70%
128 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.46M req/s	1.54M req/s	3.52%
2048 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.04M req/s	1.04M req/s	-0.000240%

GET - P1, IO 9



Name	Min	Max ~	Difference percent
16 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.50M req/s	1.69M req/s	10.3%
96 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.45M req/s	1.68M req/s	14.0%
128 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.48M req/s	1.64M req/s	6.59%
2048 Bytes, 9 IO-Threads, 1 Pipelining, aarch64	1.04M req/s	1.04M req/s	0.000926%

04

活跃的全球与本土社区

Valkey leadership

Project Technical Steering Committee

 Binbin Zhu Tencent  	 Harkrishn Patro AWS  	 Jacob Murphy Google Cloud  
 Lucas Yang Percona  	 Madelyn Olson Committee Chair AWS  	 Ping Xie Oracle  
 Ran Shidlansik AWS  	 Viktor Söderqvist Ericsson  	 Zhao Zhao Alibaba  

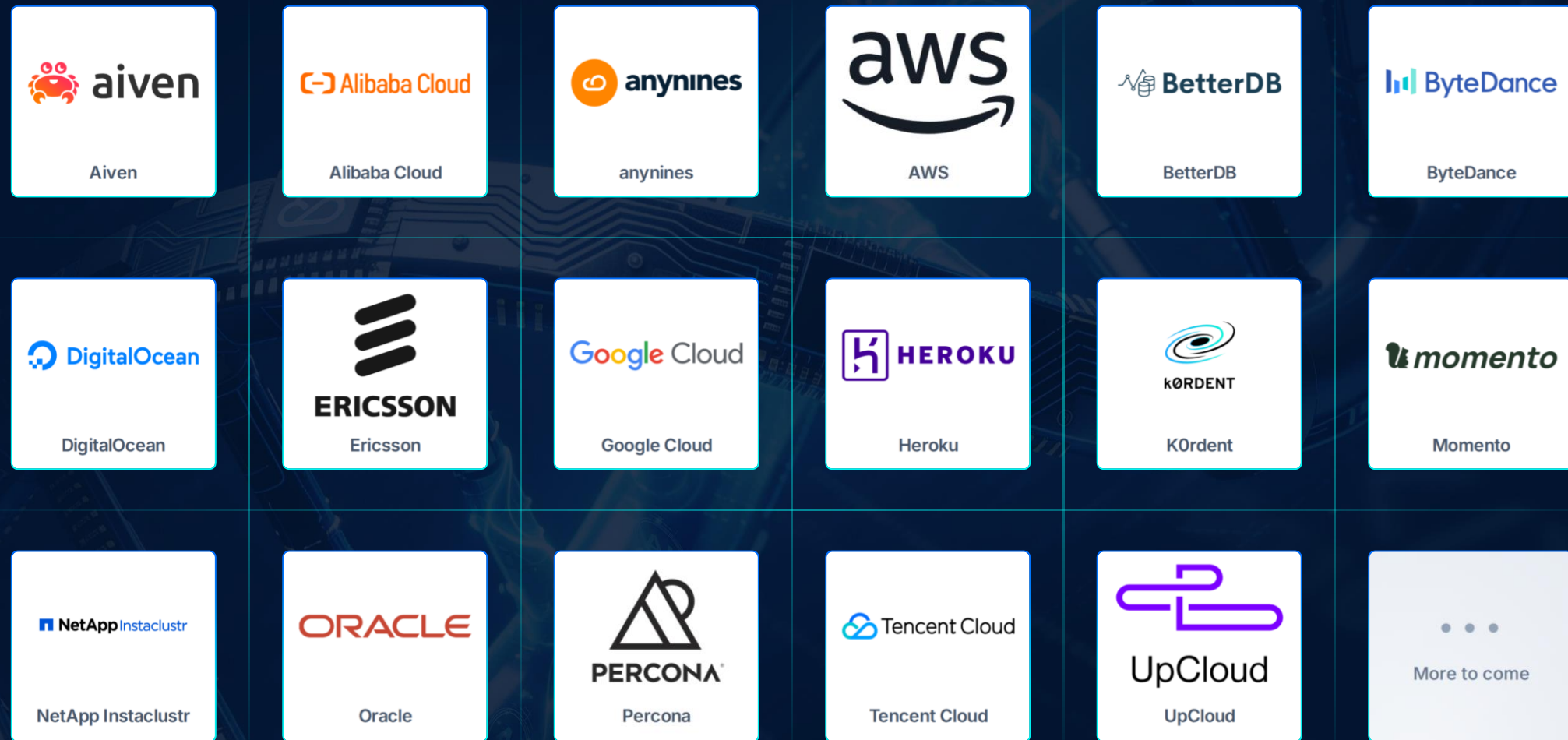
多元化的贡献者

Contributors



kurtmckee, twooster, and 82 other contributors

Participate with Valkey



<https://valkey.io/participants/>

腾讯云率先在国内支持 Valkey



Valkey offering:

[Tencent Cloud Distributed Cache \(English\)](#) is a caching and storage service developed by Tencent Cloud, compatible with Redis® OSS, Valkey, and Memcached engines. Its rich data structures help you develop various business scenarios. It supports primary-replica hot standby and provides a full suite of database services including automatic disaster recovery failover, data backup, fault migration, instance monitoring, online scaling, and data rollback.

About the organization:

[Tencent Cloud \(English\)](#), one of the world's leading cloud companies, is committed to creating innovative solutions to resolve real-world issues and enabling digital transformation for smart industries. Through our extensive global infrastructure, Tencent Cloud provides businesses across the globe with stable and secure industry-leading cloud products and services, leveraging technological advancements such as cloud computing, Big Data analytics, AI, communication and network security.

www.cloud.tencent.com/product/tcdc
www.tencentcloud.com/products/tcdc



Madelyn Olson • 已关注

Valkey project maintainer and Principal Engineer for AWS in-memory...

3 个月前 • 🌐

I'm always excited to see more clouds offering [#Valkey](#), but I think it's more exciting to highlight the incredible work BinBin has done on the project. His contributions have reduced failover times, fixed race conditions, and made Valkey simpler to operate at scale by building the foundation for atomic slot foundation. I'm honored to continue to continue to work with him with the support of Tencent cloud!

显示译文



Tencent Cloud

33,799 位关注者

3 个月前 • 🌐

We are proud to announce that Tencent Cloud now officially supports [#Valkey](#) 8.0 as a leading Asian cloud provider — delivering a production-grade, open alternative for Redis users.

Our NoSQL expert Binbin Zhu has become one of the most active core contributors to the [#Valkey](#) community over the past two years, with 359 commits, ranking #1 globally.

You come for the software, you stay for the community



Valkey Keyspace 2025 Amsterdam

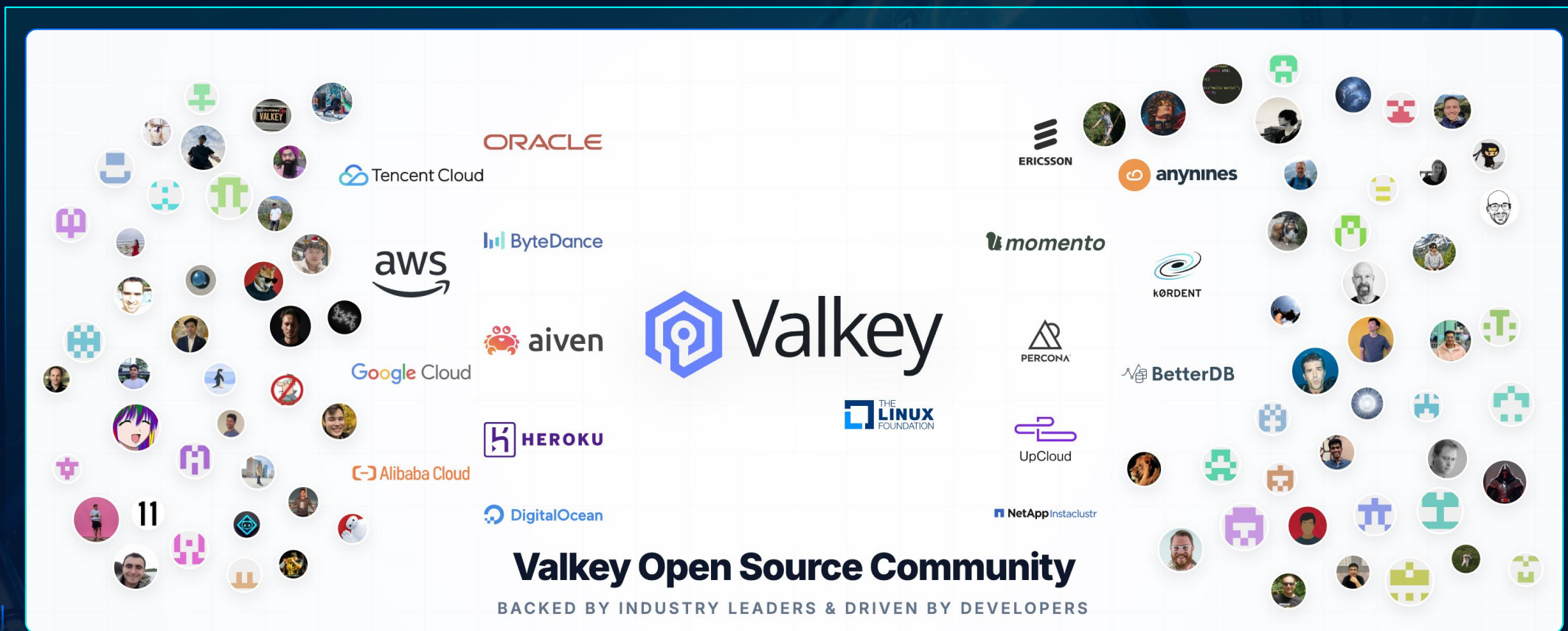


Valkey Keyspace 2025 Beijing

05

加入 Valkey

我们共同构建了什么？



How to Participate in and Contribute to Valkey

How to Contribute with us?

We welcome your involvement in the Valkey community! Here are several ways you can contribute:



Ask Questions

If you have any inquiries about Valkey, feel free to join the conversation on our GitHub discussions or chat with us on Slack.

[GitHub Repository](#)

[Slack](#)



Report Bugs

If you encounter any issues while using Valkey, please help us improve the project by filing a bug report at our GitHub repository.

[GitHub Repository](#)



Connect on Social Media

Stay updated and connect with us on our social media platforms.

[LinkedIn](#)

[Bluesky](#)

[Twitter](#)



Suggest Features

We value your ideas! If you have a suggestion for a new feature, please submit a feature request on our GitHub.

[GitHub](#)



Security Concerns

For any potential security issues, please refer to our Security Policy.

[Learn more](#)



Community Conduct

If you experience any issues with community members behavior, kindly check our Code of Conduct for guidance.

[Learn more](#)

| What's coming next for Valkey

**Read command
offloading**

**New multiplexing
protocol**

Forkless replication

Raft based Cluster V2

B+ tree Sorted Set

Full sync from replica

Active Active

<https://github.com/orgs/valkey-io/projects/51/views/1>

| Enjoy Valkey



No CLA, only have DCO



<https://github.com/valkey-io/valkey>



<https://github.com/valkey-io/valkey-doc>



<https://github.com/valkey-io/valkey/blob/unstable/CONTRIBUTING.md>

THANKS

